

Name: _____ Gabrielle D., Ethan R., Tez, Gabrielle S. _____

Date: _____

Class: _____

MyDesign Portfolio

Element C Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to present your design requirements, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own design requirements, be sure to delete the instructions. Make sure to list your design requirements in ranked order, with design requirement #1 having the highest priority. If possible, list 5 requirements that are labeled as HIGH priority and 1-5 additional requirements that are labeled as LOW priority.

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 Collected Water/Not Municipal (High Priority)

It is a requirement that the design must use collected water unless there is drought, municipal water should only be used to supplement when there is a lack of water from rain.

This requirement ties to our research because we researched how to collect rainwater and the weather/rain patterns in the area in which the problem occurs.

The Stakeholder (Dr.Jackson) said we could use municipal water if needed to supplement. The water collection system cannot be directly hooked up to the municipal water supply, even if it only turns on when supplementing.

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Design Requirement #2 Transportable and Above Ground Irrigation System (High Priority)

The design must be above ground and transportable. The client does not want the irrigation system to be dug in the ground or any part of the system underground. It must be able to be moved and picked up when needed.

The requirement for the system being above ground leads to looking into above ground hoses and an above ground collection tank. Our prior research was confined to above ground systems and finding what works best for watering plants with no underground sprinklers but open and movable sprinklers.

The design must be movable so if the client shifts the location of the plants in the garden then the client will also be able to move the irrigation system with the plants in the garden.

Design Requirement #3: Sustainable Water System (High Priority)

The irrigation system should be able to support itself individually for at least a week, while the owner is gone.

The irrigation system will require a water tank that is able to collect natural water sources, such as rain, and also should be able to apply water to 12x12 gardens in a timely manner that will not kill the plants.

Based on research the system will be able to collect water from the gutter of the roof, leaving an opening in the tank provided by the primary stakeholder to collect rainwater. Then using a water hose and a system that uses a time based function, we will use it to water plants.

Design Requirement #4: Budget is \$100 (Mid Priority) (budget can change/adapt)

It is a requirement that our budget must be \$100.

Our research shows that irrigation systems can be very expensive. However, we only have \$100 to spend on supplies to design our irrigation system.

Dr. Jackson informed us that we only have \$100 to spend. He thinks that this is an adequate amount of money to spend on an above-ground irrigation system. Our irrigation system will not be too expensive.